(Task 1 of 4) In this tutorial, we will review previously learned content. 0

Lispy |

(Task 2 of 4) What is the result of running this program?

Lispy [Run ▶]

```
(defvar i (* j 3))
(defvar j 2)

i
j
```

Scala 3

```
val i = j * 3
val j = 2
println(i)
println(j)
```

`error` 2

You predicted the output correctly 🎉🎉🎉 3

The first definition tries to bind `i` to the value of `(* j 3)`. To evaluate `(* j 3)`, we need the value of `j`. But `j` is not bound to a value at that moment.

Click here to run this program in the Stacker.

Lispy |

(Task 3 of 4) What is the result of running this program?

Lispy [Run ▶]

```
(defvar a 3)
(deffun (foo b)
  (deffun (bar)
    (defvar c 6)
    (+ a b c))
  (bar))
(+ (foo 4) 2)
```

JavaScript

```
let a = 3;
function foo(b) {
  function bar() {
    let c = 6;
    return a + b + c;
  }
  return bar();
}
console.log(foo(4) + 2);
```

`15` 5

You predicted the output correctly 🎉🎉🎉 6

This program binds `a` to `3` and `foo` to a function, and then evaluates `(+ (foo 4) 2)`. The value of `(+ (foo 4) 2)` is the value of `(+ (bar) 2)`, which is the value of `(+ (+ a b z) 2)`, which is the value of `(+ (+ 3 4 6) 2)`, which is `15`.

Click here to run this program in the Stacker.

Lispy |

(Task 4 of 4) What is the result of running this program?

Lispy [Run ▶]

```
(deffun (k a)
  (defvar b 1)
  (+ a b))

(+ (k 3) b)
```

Scala 3

```
def k(a : Int) =
  val b = 1
  a + b
println(k(3) + b)
```

`error` 8

9

You predicted the output correctly 🎉🎉🎉

`(+ (k 3) b)` is evaluated in the top-level block, where `b` is not defined. So, this program errors.

Click here to run this program in the Stacker.

You have finished this tutorial 🎉🎉🎉

Please print the finished tutorial to a PDF file so you can review the content in the future. **Your instructor (if any) might require you to submit the PDF.**

Start time: 1781414184707